



BSc (Hons) in **Information Technology**
Specialized in AI
IT3012 : Intelligent Agents
Year 3 · Semester 1 · 2026 Semester 2

Faculty of Computing

Practical 04

Module Objective: In previous weeks, your Drow Ranger agent used A* Search to find a physical path to the goal. However, it blindly assumed every non-wall tile was safe. Today, we implement a **Knowledge Base (KB)** and a **Forward Chaining Inference Engine**. Your agent must now mathematically prove a cell is logically *Feasible* before taking a step, using real-time rules about map vision and enemy threats.

Part 1: The Logic Engine Architecture

Step 1.1: Building the Knowledge Base (KB)

Instead of hardcoding hundreds of nested `if-else` statements in our agent's movement logic, we will build a declarative Knowledge Base. You need to design a Python class that stores **Facts** (current percepts) and **Rules** (Horn Clauses).

- 1. Create a new file named `logic_engine.py` .
- 2. Implement a `KnowledgeBase` class based on the following structural requirements:

```
CLASS KnowledgeBase:
    // Attributes
    DECLARE facts AS Set           // To store unique string fact
    DECLARE rules AS List         // To store rules as Tuples: (

    // Methods
    FUNCTION tell_fact(fact_string):
        ADD fact_string TO facts

    FUNCTION tell_rule(premise_list, conclusion_string):
        APPEND Tuple(premise_list, conclusion_string) TO rules

    FUNCTION clear_facts():
        EMPTY the facts Set
```

Part 2: Implementing Forward Chaining

Step 2.1: The Inference Algorithm

The agent needs an Inference Engine to deduce new information from its raw sensors. Translate the following Data-Driven **Forward Chaining** algorithm into a Python method inside your `KnowledgeBase` class.

- 1. Write the `forward_chain()` method.
- 2. Ensure you use a `while` loop that continues to iterate over the rules until a full pass results in absolutely no new facts being deduced.

```
FUNCTION forward_chain():
    DECLARE new_facts_added = TRUE

    WHILE new_facts_added == TRUE:
        new_facts_added = FALSE

        FOR EACH (premises, conclusion) IN rules:
            IF conclusion IS NOT IN facts:

                // Modus Ponens Check
                IF ALL premises ARE IN facts:
                    ADD conclusion TO facts
                    new_facts_added = TRUE
```

Part 3: Hooking Logic into the Grid Game

Step 3.1: Defining the Game Constraints

Now we translate the game's safety constraints into our logic engine using Propositional Logic.

- 1. In your main `agent.py`, instantiate the KB in the `__init__` method: `self.kb = KnowledgeBase()`.
- 2. Use your `tell_rule` method to define the following safety rules:
`Rule 1: TargetVisible  $\wedge$  HasDust  $\Rightarrow$  SafeToEngage`
`Rule 2: SafeToEngage  $\wedge$  BloodseekerMissing  $\Rightarrow$  Retreat`

Step 3.2: Validating Feasibility in A*

Modify your existing A* pathfinding algorithm to consult the Knowledge Base before expanding a node.

- 1. Locate the neighbor-generation loop in your A* code (where you normally check for physical walls).
- 2. Before adding a neighbor to the `open_list`, clear the KB facts.
- 3. Feed the current percepts for that specific tile into the KB using `tell_fact()`.

4. Run `self.kb.forward_chain()` .
5. If `'Retreat'` is deduced and exists in `self.kb.facts` , mark the tile as **Infeasible**. Skip it (do not add it to the open list), even if it is physically reachable.

Part 4: Self-Evaluation Test Cases

4.1 Automated Validation

Create a file named `test_logic.py` . Copy and run the following Python assert statements to verify your Forward Chaining engine works correctly mathematically before you attempt to integrate it into the visual grid.

```
from logic_engine import KnowledgeBase

def test_forward_chaining():
    kb = KnowledgeBase()

    # Add Domain Rules
    kb.tell_rule(['TargetVisible', 'HasDust'], 'SafeToEngage')
    kb.tell_rule(['SafeToEngage', 'BloodseekerMissing'], 'Retreat')

    # Test Case 1: Safe Engagement
    kb.clear_facts()
    kb.tell_fact('TargetVisible')
    kb.tell_fact('HasDust')
    kb.forward_chain()
    assert 'SafeToEngage' in kb.facts, "Test 1 Failed: Should deduce Sa
    assert 'Retreat' not in kb.facts, "Test 1 Failed: Should NOT deduce

    # Test Case 2: Unsafe Engagement (Bloodseeker Missing)
    kb.clear_facts()
    kb.tell_fact('TargetVisible')
    kb.tell_fact('HasDust')
    kb.tell_fact('BloodseekerMissing')
    kb.forward_chain()
    assert 'Retreat' in kb.facts, "Test 2 Failed: Should deduce Retreat

    print("✅ All Logic Engine Test Cases Passed!")

if __name__ == "__main__":
    test_forward_chaining()
```

Part 5: Theory-to-Implementation Mapping

Based on the code you just wrote and the lecture on Logical Reasoning, complete the following analytical questions to explain *how* your code implements the theory.

1. Reachability vs. Feasibility (Apply)

In Step 3.2, you modified the A* algorithm's neighbor-generation loop. How does your new code differentiate between checking if a node is "Reachable" (like a wall collision) versus checking if it is "Feasible"? Explain how the Knowledge Base enforces feasibility.

Reachability checks whether the agent can physically move to a node, such as checking grid boundaries and walls.

Feasibility checks whether moving to that reachable node is logically acceptable. The Knowledge Base uses facts and rules with forward chaining. If it deduces Retreat, the node is considered infeasible and A* skips it.

2. Declarative vs. Procedural Paradigms (Analyze)

Why did we build a `KnowledgeBase` class with a `forward_chain()` loop instead of simply writing `if 'TargetVisible' in percepts and 'HasDust' in percepts:` directly inside the agent's movement code? What happens to the agent's code complexity if we add 50 new game rules?

A KnowledgeBase keeps the logical rules separate from the agent's movement code. This makes the program easier to understand, maintain, and extend.

If we used many if-else statements, adding 50 new rules would make the agent code very complex. With the KB, new rules can simply be added without changing the main A* logic.

3. Modus Ponens & Horn Clauses (Understand)

The rule `['TargetVisible', 'HasDust'] ⇒ 'SafeToEngage'` is formally known as a Horn Clause. Briefly explain how your Python implementation of the `forward_chain()` `while` loop acts as a mathematical execution of the *Modus Ponens* inference rule.

The rule:

$\text{TargetVisible} \wedge \text{HasDust} \Rightarrow \text{SafeToEngage}$

is a Horn Clause.

The `forward_chain()` method checks whether all the premises exist in the facts. If both are true, it adds `SafeToEngage` as a new fact.

This implements **Modus Ponens**: when the required premises are true, the conclusion can be inferred.

Finalize and Lock Lab 05 Implementation



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